ARCHITECTURE GOVERNANCE & DECISION LOG

LoanLens System Architecture Records

Formal Architecture Decision Records (ADRs 001–008) governing regulatory compliance, dual-mode deployment, cryptographic audit ledgers, deterministic credit math, and advisory RAG integration.

TARGET SYSTEM	STANDARD & VERSION	REGULATORY SCOPE	STATUS
LoanLens Platform	**MADR 3.0.0 (Rev 2026.11)**	**Maker-Checker & Auditability**	**All Approved / Enforced**

SYSTEM INVARIANT CHARTER

LoanLens enforces four non-negotiable architectural invariants: **(1) Zero-API Client Capability** for friction-free portfolio inspection; **(2) Maker-Checker RBAC** establishing cryptographic segregation between Loan Officers and Reviewers; **(3) Deterministic Credit Math** barring non-deterministic LLMs from loan approval sizing; and **(4) Immutable Ledger Cryptography** preserving audit lineage via sequential SHA-256 block chaining.

ADR ID	DECISION SUBJECT	DOMAIN / SUBSYSTEM	STATUS
ADR-001	Dual-Mode Architecture (Client Showcase vs. Live API)	Deployment & Runtime	ACCEPTED
ADR-002	Maker-Checker Role Separation & RBAC State Machine	Security & Governance	ACCEPTED
ADR-003	Deterministic FOIR & Multi-Tenor Debt Engine	Credit Risk & Underwriting	ACCEPTED
ADR-004	Cryptographic SHA-256 Tamper-Evident Audit Ledger	Compliance & Data Integrity	ACCEPTED
ADR-005	Policy-Grounded Advisory RAG with Sliding-Window Chunking	AI Architecture & LLM Safety	ACCEPTED
ADR-006	Lightweight Single-Node Worker via PostgreSQL Polling	Infrastructure & Queueing	ACCEPTED
ADR-007	Immutable Record Retention & Audited "Discard" Lifecycle	Regulatory Compliance	ACCEPTED
ADR-008	Dual-Stream Income Accounting (Gross vs. Net Cash Flow)	Underwriting Domain Model	ACCEPTED

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ADR-001 • ARCHITECTURE / DEPLOYMENT

Dual-Mode Architecture: Standalone Showcase & Live Full-Stack API

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Portfolio evaluators, hiring managers, and recruiters typically dedicate less than two minutes reviewing an application. Forcing external users to execute Docker containers, provision PostgreSQL, and establish Redis brokers introduces severe friction that drops evaluation rates to near zero. Concurrently, the platform must demonstrate production-grade full-stack backend maturity.

DECISION

Implement a runtime dual-mode abstraction controlled via `VITE_DEMO_MODE=true/false`. When enabled, the SPA delegates to an in-memory client repository reproducing RBAC workflows, SHA-256 ledger chaining, and in-browser RAG retrieval. When disabled, the frontend routes network calls to FastAPI endpoints over `/api/v1`.

POSITIVE CONSEQUENCES

- Zero hosting cost and instantaneous load times via Vercel Edge CDN.
- No external database dependencies or server outages for evaluators.
- Single codebase shares domain models, calculations, and UI components.

TRADE-OFFS & MITIGATIONS

- Repository layer requires twin implementations (Live vs Mock).
- Mitigated via strict TypeScript interfaces matching FastAPI Pydantic schemas.

ADR-002 • SECURITY & REGULATORY GOVERNANCE

Maker-Checker Role Separation (Four-Eyes Principle)

ACCEPTED

CONTEXT & PROBLEM STATEMENT

In retail and commercial lending, regulatory frameworks mandate segregation of duties (the "Four-Eyes Principle"). The party entering loan application parameters ("Maker") must never be granted unilateral authorization to approve credit lines ("Checker"), eliminating internal collusion and operational fraud.

DECISION

Enforce strict role-based access control (RBAC) across three distinct personas: `Loan Officer` (Maker: creates/edits/submits drafts), `Underwriting Reviewer` (Checker: evaluates risk, issues terminal Approve/Reject dispositions), and `Admin` (Governance: inspects audit ledgers, verifies hash chains). Transitions between states are guarded by a deterministic finite state machine.

POSITIVE CONSEQUENCES

- Guarantees strict auditability conforming to central banking regulations.
- UI dynamically adapts permissions, suppressing approval actions for Officers.

TRADE-OFFS & MITIGATIONS

- Requires persona simulation switcher in showcase mode for user exploration.

ADR-003 • RISK CALCULATION ENGINE

Deterministic FOIR Engine with Dynamic Tenor Selection

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Credit underwriting relies on the Fixed Obligation to Income Ratio (FOIR). Relying on machine learning or generative models for arithmetic calculations introduces stochastic hallucination risks that violate statutory lending mandates. Furthermore, hardcoded loan repayment periods fail real-world repayment structuring.

DECISION

Calculate all EMI and FOIR metrics through a clean-room deterministic engine using standard actuarial equations: `EMI = [P × r × (1+r)^n] / [(1+r)^n − 1]`. Dynamic tenors (12, 24, 36, 48, 60 months) are parameterized at intake, with net monthly take-home serving as the non-negotiable cash-flow divisor.

POSITIVE CONSEQUENCES

- 100% reproducible credit math with zero risk of arithmetic drift.
- Instant recalculation in UI as underwriters adjust requested tenors or amounts.

TRADE-OFFS & MITIGATIONS

- Complex amortizations must be synchronized between TypeScript and Python.

ADR-004 • COMPLIANCE & SECURITY

Tamper-Evident SHA-256 Sequential Audit Ledger

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Audit trails in standard relational schemas are vulnerable to silent retroactive updates or unauthorized SQL deletions by database administrators, exposing financial institutions to insider fraud and severe compliance penalties.

DECISION

Store all domain events in an append-only cryptographic ledger. Each block computes a SHA-256 digest binding the prior block hash, timestamp, actor ID, state payload, and transition type: `H_n = SHA-256(H_{n-1} || Timestamp || Actor || Payload)`. The Admin console includes an automated verification algorithm, mutation simulator, and repair tool.

POSITIVE CONSEQUENCES

- Any retroactively modified record immediately breaks subsequent hashes.
- Demonstrates enterprise cryptography running natively across both client and server.

TRADE-OFFS & MITIGATIONS

- Requires append-only table structure; updates and hard deletes are banned.

ADR-005 • AI ARCHITECTURE & SAFETY

Advisory-Only RAG with In-Browser Document Parsing

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Credit underwriters must consult complex, frequently revised internal circulars and regulatory guidelines (e.g., RBI notifications). Directing queries to public general LLMs produces ungrounded citations and risks leaking confidential applicant data.

DECISION

Implement an advisory-only Retrieval-Augmented Generation (RAG) copilot. The system parses uploaded `.pdf`, `.docx`, and `.txt` documents in the browser using `pdfjs-dist` and `mammoth`, indexes them via 350-word sliding windows, performs semantic/cosine scoring, and synthesizes cited recommendations with a prominent guardrail notice.

MANDATORY GUARDRAIL: AI outputs are strictly advisory. The copilot is architecturally barred from writing directly to application decision columns without explicit human confirmation.

POSITIVE CONSEQUENCES

- Grounded clause citations with exact similarity percentages and document provenance.
- Operates locally in client memory with zero external API bill for demo visitors.

TRADE-OFFS & MITIGATIONS

- Client-side parsing limited to files under 25MB to prevent memory exhaustion.

ADR-006 • QUEUEING & INFRASTRUCTURE

PostgreSQL-Backed Worker Queue for Single-Node Simplicity

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Background tasks (virus scans via ClamAV, PDF generation, asynchronous bureau checks) initially targeted Celery + Redis. However, running Redis alongside Celery workers requires multi-container orchestration that complicates single-node setups and increases minimum VPS RAM requirements.

DECISION

Defer Celery/Redis in default deployment profiles in favor of a lightweight PostgreSQL row-locking queue (`SELECT ... FOR UPDATE SKIP LOCKED`). This satisfies asynchronous processing guarantees within the existing relational database without spinning up external message brokers.

POSITIVE CONSEQUENCES

- Reduces container footprint and operational complexity by 50%.
- Atomic transactions: job state transitions commit inside database transactions.

TRADE-OFFS & MITIGATIONS

- Max throughput bounded by DB I/O (sufficient for thousands of loans per hour).

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ADR-007 • COMPLIANCE & RETENTION

Immutable Record Retention & Audited "Discard" Lifecycle

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Users frequently expect a "Delete Application" button. However, in banking environments, hard-deleting records violates statutory recordkeeping rules and breaks ledger chain continuity.

DECISION

Ban physical SQL `DELETE` operations across all application tables. Draft applications can only transition to a soft-archived `Discarded` state, accompanied by an explicit `DRAFT_DISCARDED` event hashed into the SHA-256 audit ledger. Non-draft states (Submitted, Approved, Rejected) cannot be discarded or removed under any circumstances.

POSITIVE CONSEQUENCES

- 100% compliance with financial retention statutes and evidentiary requirements.
- The SHA-256 ledger maintains unbroken cryptographic chain continuity.

TRADE-OFFS & MITIGATIONS

- Requires filter controls in UI dashboards to separate active and discarded loans.

ADR-008 • FINANCIAL MODELING

Dual-Stream Income Accounting: Gross Earnings vs. Net Cash Flow

ACCEPTED

CONTEXT & PROBLEM STATEMENT

Lumping all earnings into a single "Income" input leads to severe underwriting errors: calculating debt ratios against gross salary overstates cash flow, while using net salary alone obscures tax deductions and employer provident fund commitments.

DECISION

Separate income intake into two distinct fields: `Monthly Gross Income` (used for tax bracket verification and income-tax return matching) and `Monthly Net Take-Home` (used strictly as the cash-flow denominator for FOIR and EMI obligations).

POSITIVE CONSEQUENCES

- Accurate debt-service evaluation reflecting actual liquid disposable earnings.
- Prevents predatory over-leveraging and non-conforming approvals.

TRADE-OFFS & MITIGATIONS

- Adds an additional field to applicant intake forms and data schemas.

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